MILITARY SPECIFICATION

HEADSET-MICROPHONE, H-46A/UR

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one type of headset and microphone combination; designated Headset-Microphone H-46A/UR.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein.

2.1.1 Specifications.-

Military

MIL-A-140	Adhesive, Water-Resistant, Waterproof Barrier-Material
MIL-M-5793	Microphone M-3A/A
JAN-P-100	Packaging and Packing for Overseas Shipment - General Specification
JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes; Wood, Nailed
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible

Air Force-Navy Aeronautical

AN-P-13	Preservation and Packaging; Parts and Equipment General Specification For
AN-B-20	Barrier; Flexible Sheet Moisture Vapor

U. S. Air Force

40985	Marking of Interior Packages (For Communications Equipment)

Bureau of Aeronautics

M-587	Mounting; Flying Helmet and Headband Earphone

2.1.2 Other Publications.-

Military Standard (Book)

MIL-STD-129	Marking of Shipments

2.1.3 Drawings.-

U. S. Air Force

SC-D-2216	Headband HB-7, Details
49C12798	Headset HS-33-A, Assembly
49B13276	Cushion - Headset Modified
49B13277	Holder - Receiver Headset Cushion
46D40562	Headset - Microphone H-46A/UR, Assembly
SC-D-5975	Carbon Microphone Test Circuits

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Navy Aircraft Factory

215466 Washer - Aviators' Helmet and Headbands Phone Unit Mounting

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Components.- The headset-microphone shall consist of the following components assembled in accordance with Drawing 46D40562.

1 Microphone M-3A/A, in accordance with Specification MIL-M-5793
1 Headset HS-33-A, in accordance with Drawing 49C12798
1 Instruction Sheet

3.1.1 Instruction Sheet.- The printed instruction sheet shall briefly describe the procedure to be followed in adjusting and using the headset-microphone. The instruction sheet shall include a note in bold type, that the microphone cover installed as part of Microphone M-3A/A is a frost and windblast shield, and is not to be removed from the microphone unit when the headset-microphone is put into use. The instruction sheet shall be approved by the procuring agency.

3.2 Earphone Cushions.- The earphone cushions shall be in accordance with Specification M-587 with the following exceptions or additions:

3.2.1 Positioning Mold Marks.- The earphone cushions shall include mold marks which locate the positions of the mounting holes for Bracket MT-522A/U, part of Microphone M-3A/A. The mold marks shall be in accordance with Drawing 49B13277.

3.2.2 Condition after Molding.- After molding, the socket shall be a firmly bound, tough, elastic and homogeneous mass; and the socket shall have a relatively smooth uniform appearance.

3.2.3 Compression.- The vertical compression of the pad resulting from the addition of a 450-gram weight as specified in figure 5 of Specification M-587 shall be not less than 0.11 inch nor more than 0.18 inch.

3.2.4 Modification of Earphone Cushion Washers.- The hole diameter of the washer covered by Drawing 215466, Part No. 215466-2, shall be 1-13/16 inches.

3.3 Government Furnished Property.- When provided for in the contract or purchase order, one Headset HS-33 will be furnished by the Government to the contractor for the fabrication of the headset-microphone.

3.3.1 Conversion.- The following procedure shall be followed to convert Headset HS-33 to Headset HS-33-A:

3.3.1.1 Remove the cushions (MC-162-A or MX-41/AR) and replace them with cushions in accordance with Drawing 49B13276.

3.3.1.2 Remove the identification tag bearing the marking "Headset HS-33" and substitute therefor an identification tag marked "Headset HS-33-A."

3.3.1.3 Incorporate the bends of figure 1, Drawing SC-D-2216, on the headset headband. It is possible to make these bends on the steel bands without disassembling the headband.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Classification of Tests.- The inspection and testing of the headset or headset-microphone shall be classified as follows:

(a) Preproduction tests: Preproduction tests are those tests accomplished on samples representative of the production of the headset-microphone after the award of the contract, to determine that the production headset-microphone meets the requirements of this specification.

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(b) Inspection tests: Inspection tests are those tests accomplished on headset-microphones manufactured and submitted for acceptance under contract.

4.2 Preproduction Tests.-

4.2.1 Sampling Instructions.- Two headset-microphones and 12 earphone cushions as specified in the contract will be tested for design approval by the procuring agency, or when so specified in the contract, at the contractor's plant under the supervision of the procuring agency.

4.2.1.1 Status of Samples.- The samples shall be, in all respects, truly representative of the equipment the contractor proposes and expects to furnish in quantity production.

4.2.1.2 Parts Data.- The preproduction samples shall be accompanied by adequate identifying information, including the following:

(a) Manufacturer of the earphone cushions.

(b) Manufacturers of any part not supplied by the prime contractor.

4.2.1.3 Certification.- The contractor shall, in writing, certify to the procuring agency that the rubber used in the cushion has been carefully compounded and vulcanized and complies with the requirements of Specification M-587. Test data on the rubber compounds are to be furnished as part of the certificate. Certificates indicating that the chamois and kapok comply with the specification requirements shall also be furnished.

4.2.2 Tests.- The Preproduction tests shall consist of a thorough visual and mechanical inspection and an operations test to determine compliance with the requirements of this specification, and also any Preproduction tests specified in subsidiary specifications which are deemed necessary to determine complete compliance with the requirements stated therein.

4.3 Inspection Tests.- The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.3.1 Individual Tests.- Each Headset-Microphone H-46A/UR shall be subjected to the following tests, as applicable.

4.3.2 The earphone cushions and Microphone M-3A/A shall be inspected in accordance with the individual test requirements of their respective specifications.

4.3.3 Examination of Product.- The headset-microphone shall be given a thorough visual inspection for compliance with the material and workmanship requirements specified herein.

4.3.4 Operation Test.- The headset-microphone shall be given a talking-listening test using the test circuit shown in figure 1, Drawing SC-D-5975. The headset shall be plugged into a Headset Adapter MC-385-( ) (or suitable equivalent transformer) which in turn is plugged into the "A-C Output" jack of the electronic voltmeter of the test circuit. During this test the caller shall listen to his own voice and insure that the Headset-Microphone H-46A/UR is performing in a normal and satisfactory manner.

5. PREPARATION FOR DELIVERY

5.1 Application.- The packaging, packing, and marking requirements specified herein apply only to direct purchases by or direct shipments to the Government.

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5.2 Packaging.- The headset or headset-microphone, as applicable, shall be either commercially packaged or unit packaged as specified by the procuring agency.

5.2.1 Commercial Packaging.- No packaging beyond that normally provided under best commercial practice will be required.

5.2.2 Unit Packaging.- Each headset-microphone shall be unit packaged in accordance with Specification AN-P-13, Method II, carton-barrier-carton. Barrier materials shall be in accordance with Specification AN-B-20. Cushioning materials shall be in accordance with Specification JAN-P-100, except that excelsior will not be acceptable.

5.3 Packing.- Each headset or headset-microphone, as applicable, shall be packed for either domestic or overseas shipment as specified by the procuring agency. Exterior containers, insofar as possible, shall be uniform in size and shape and shall contain an identical number of like items only and shall be designed to enclose the contents in a compact manner.

5.3.1 Domestic Packing.- Domestic type exterior containers shall conform to the requirements of Consolidated Freight Classification Rules in effect at the time of shipment and shall be designed to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. Where fiberboard is used, it shall have a minimum Mullen test of 275 pounds.

5.3.2 Overseas Packing.- Overseas type exterior containers shall be cleated plywood boxes in accordance with Specification JAN-P-105 and plywood shall be type B, condition I. As an alternative, nailed wood boxes in accordance with Specification JAN-P-106, style 2 or 4 may be used. In either event, the box shall be lined with a waterproof case liner of material in accordance with Specification JAN-P-125, sealed with material in accordance with Specification MIL-A-140.

5.4 Marking and Labeling.-

5.4.1 Packages.- Interior packages shall be marked in accordance with Specification 40985.

5.4.2 Shipping Containers.- Each shipping container shall be marked in accordance with the requirements applicable to the individual Services, as specified in MIL-STD-129.

6. NOTES

6.1 Intended Use.- Headset-Microphone H-46A/UR is a headset and microphone combination intended to replace Headset HS-33 and Microphone T-30-( ) in Military aircraft for all altitudes up to 15,000 feet.

6.2 Ordering Data.- Invitations for bids, contracts and purchase orders should state the conditions for the following.

6.2.1 Preproduction Tests.- It is expected that the contract or purchase order will specify that two headset-microphones and 12 earphone covers will be required as First Articles (or preproduction samples) and that these First Articles (or preproduction samples) will be subjected to the Preproduction tests to determine compliance with the requirements of the specification. The invitation for bids and the contract should specify the point of inspection for these tests.

6.3 Earphone Cushions.- Specification M-587 refers to "earphone cushions" as "earphone mountings."

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6.4 Headset HS-33 furnished to the contractor for conversion into Headset HS-33A shall be new items which have never been removed from their original individual cartons. Reconditioned Headset HS-33 will not be used for this purpose.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Air Force

Other interest:
Army - OSig
Navy - Sh

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-6279

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ATTN: AFLC (SGEBA)
WRIGHT-PATTERSON AFB, OHIO 45433

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